Se-Sp_analysis

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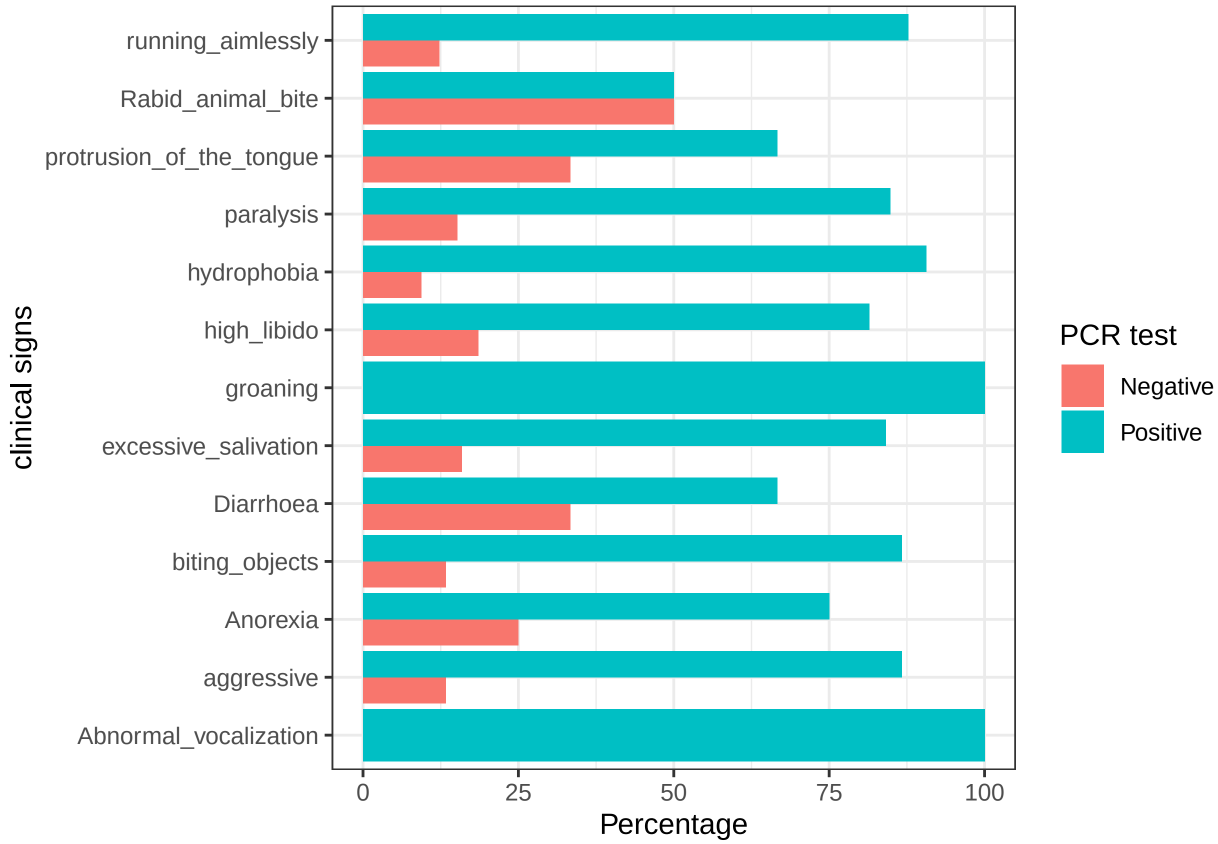
Se, Sp, NPV, PPV

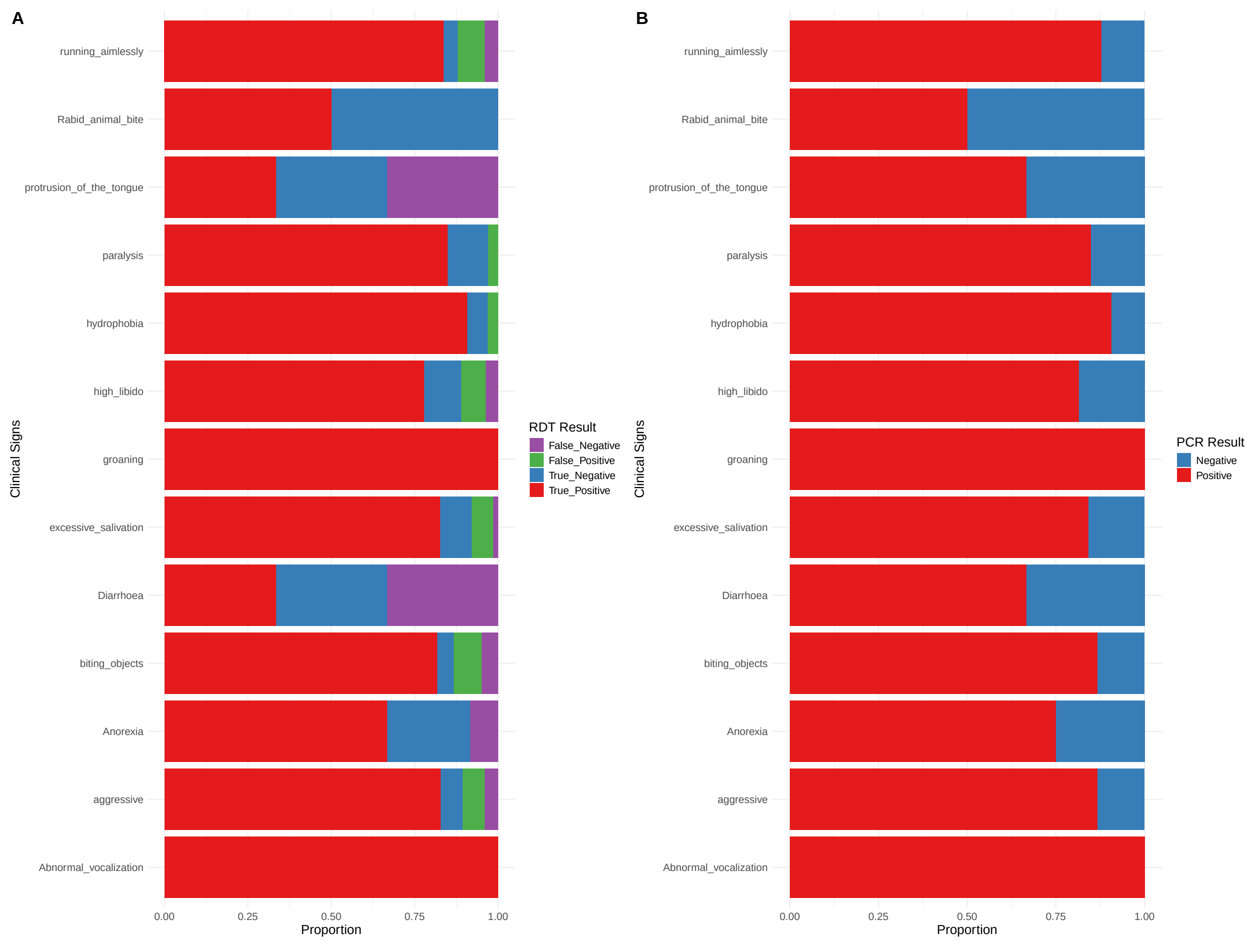
```
##           Outcome +    Outcome -      Total
## Test +           90            2         92
## Test -            6           14         20
## Total            96           16        112
##
## Point estimates and 95% CIs:
## ----------------------------------------------------------------
## Apparent prevalence *                  0.82 (0.74, 0.89)
## True prevalence *                      0.86 (0.78, 0.92)
## Sensitivity *                          0.94 (0.87, 0.98)
## Specificity *                          0.88 (0.62, 0.98)
## Positive predictive value *            0.98 (0.92, 1.00)
## Negative predictive value *            0.70 (0.46, 0.88)
## Positive likelihood ratio              7.50 (2.05, 27.45)
## Negative likelihood ratio              0.07 (0.03, 0.16)
## False T+ proportion for true D- *      0.12 (0.02, 0.38)
## False T- proportion for true D+ *      0.06 (0.02, 0.13)
## False T+ proportion for T+ *           0.02 (0.00, 0.08)
## False T- proportion for T- *           0.30 (0.12, 0.54)
## Correctly classified proportion *      0.93 (0.86, 0.97)
## ----------------------------------------------------------------
## * Exact CIs
```

PCR +VE and -VE of the relevant variables

```
##
##    Abnormal_vocalization aggressive Anorexia biting_objects Diarrhoea
##   0                    0         10        3              8         1
##   1                    4         65        9             52         2
##
##    excessive_salivation groaning high_libido hydrophobia paralysis
##   0                   10        0           5           3         5
##   1                   53        2          22          29        28
##
##    protrusion_of_the_tongue Rabid_animal_bite running_aimlessly
##   0                        1                 1                 6
##   1                        2                 1                43
```




```
## Saving 20 x 15 in image
## Saving 20 x 15 in image
```




```
## Saving 20 x 15 in image
```

Table species vs the symptoms

species	Anorexia	Diarrhoea	aggressive	biting_objects	excessive_salivation	groaning	high_libido	hydrophobia	pa
Dog	3	3	58	48	44	2	18	17	
Goat	6	0	11	11	5	0	5	4	
Cattle	3	0	5	0	13	0	3	10	
Sheep	0	0	1	1	1	0	1	1	
Total	**12**	**3**	**75**	**60**	**63**	**2**	**27**	**32**	

GLM

```
##
## Call:
## glm(formula = lab_diagnostics_pcr ~ aggressive, family = "binomial",
```

```
##     data = model_glm)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   1.5963     0.5118   3.119  0.00181 **
## aggressive    0.1906     0.4801   0.397  0.69141
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 68.275  on 81  degrees of freedom
## Residual deviance: 68.099  on 80  degrees of freedom
## AIC: 72.099
##
## Number of Fisher Scoring iterations: 4
```

```
##
## Call:
## glm(formula = lab_diagnostics_pcr ~ aggressive * species2, family = "binomial",
##     data = model_glm)
##
## Coefficients:
##                              Estimate Std. Error z value Pr(>|z|)
## (Intercept)                    2.0155     0.6365   3.166  0.00154 **
## aggressive                    -0.1010     0.4347  -0.232  0.81618
## species2Ruminants             -1.0347     0.9293  -1.113  0.26552
## aggressive:species2Ruminants   1.1351     1.1018   1.030  0.30289
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 68.275  on 81  degrees of freedom
## Residual deviance: 66.809  on 78  degrees of freedom
## AIC: 74.809
##
## Number of Fisher Scoring iterations: 4
```

```
##
## Call:
## glm(formula = lab_diagnostics_pcr ~ biting_objects, family = "binomial",
##     data = model_glm)
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept)      1.6722     0.4285   3.903 9.51e-05 ***
## biting_objects   0.1301     0.4337   0.300    0.764
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 68.275  on 81  degrees of freedom
```

```
## Residual deviance: 68.177  on 80  degrees of freedom
## AIC: 72.177
##
## Number of Fisher Scoring iterations: 4
```

```
##
## Call:
## glm(formula = lab_diagnostics_pcr ~ excessive_salivation, family = binomial(link = "logit"),
##     data = model_glm)
##
## Coefficients:
##                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)            1.8463     0.4242   4.353 1.35e-05 ***
## excessive_salivation  -0.1043     0.3501  -0.298    0.766
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 68.275  on 81  degrees of freedom
## Residual deviance: 68.191  on 80  degrees of freedom
## AIC: 72.191
##
## Number of Fisher Scoring iterations: 4
```

```
##
## Call:
## glm(formula = lab_diagnostics_pcr ~ running_aimlessly, family = binomial(link = "logit"),
##     data = model_glm)
##
## Coefficients:
##                   Estimate Std. Error z value Pr(>|z|)
## (Intercept)         1.5992     0.4119   3.882 0.000104 ***
## running_aimlessly   0.2959     0.5201   0.569 0.569330
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 68.275  on 81  degrees of freedom
## Residual deviance: 67.936  on 80  degrees of freedom
## AIC: 71.936
##
## Number of Fisher Scoring iterations: 4
```

```
##
## Call:
## glm(formula = lab_diagnostics_pcr ~ biting_objects + aggressive,
##     family = "binomial", data = model_glm)
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept)      1.5943     0.5170   3.084  0.00204 **
```

```
## biting_objects   0.0207     0.5988   0.035  0.97242
## aggressive       0.1764     0.6319   0.279  0.78011
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 68.275  on 81  degrees of freedom
## Residual deviance: 68.098  on 79  degrees of freedom
## AIC: 74.098
##
## Number of Fisher Scoring iterations: 4
```

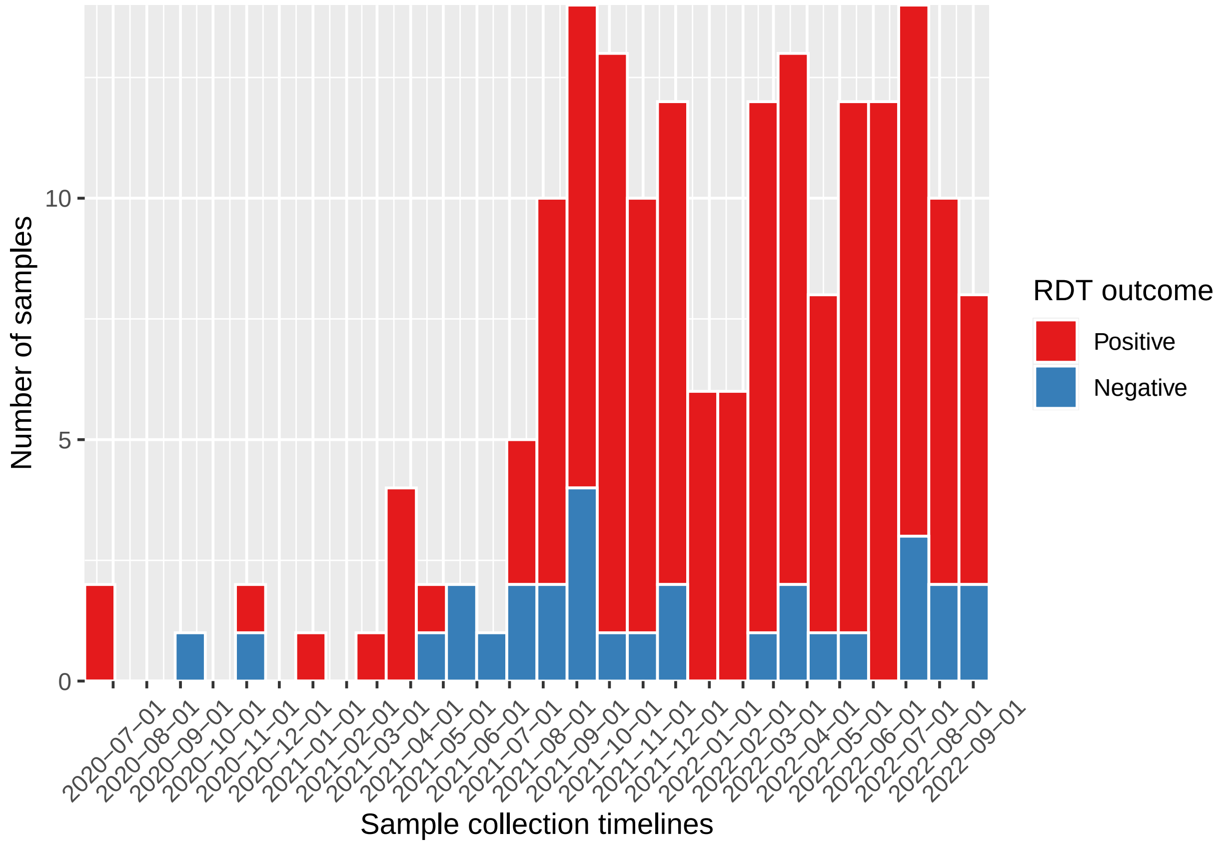
```
##
## Call:
## glm(formula = lab_diagnostics_pcr ~ aggressive + biting_objects +
##     excessive_salivation + hydrophobia + running_aimlessly +
##     high_libido + Rabid_animal_bite + paralysis + Anorexia +
##     Diarrhoea + groaning + Abnormal_vocalization, family = binomial(link = "logit"),
##     data = model_glm)
##
## Coefficients:
##                        Estimate Std. Error z value Pr(>|z|)
## (Intercept)             2.11995    0.94034   2.254   0.0242 *
## aggressive              0.67641    0.86904   0.778   0.4364
## biting_objects          0.10472    0.69549   0.151   0.8803
## excessive_salivation   -0.82736    0.66479  -1.245   0.2133
## hydrophobia             1.02522    0.75779   1.353   0.1761
## running_aimlessly       0.07981    0.74894   0.107   0.9151
## high_libido            -0.99216    0.75354  -1.317   0.1880
## Rabid_animal_bite      -3.02312    1.84087  -1.642   0.1005
## paralysis              -0.18869    0.77907  -0.242   0.8086
## Anorexia               -0.99365    0.99973  -0.994   0.3203
## Diarrhoea              -1.29409    1.64711  -0.786   0.4321
## groaning                7.91175 1978.09028   0.004   0.9968
## Abnormal_vocalization   15.66569 1971.79996   0.008   0.9937
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 68.275  on 81  degrees of freedom
## Residual deviance: 58.563  on 69  degrees of freedom
## AIC: 84.563
##
## Number of Fisher Scoring iterations: 16
```

```
##
## Call:
## glm(formula = lab_diagnostics_pcr ~ aggressive * biting_objects *
##     excessive_salivation, family = binomial(link = "logit"),
##     data = model_glm)
##
## Coefficients:
```

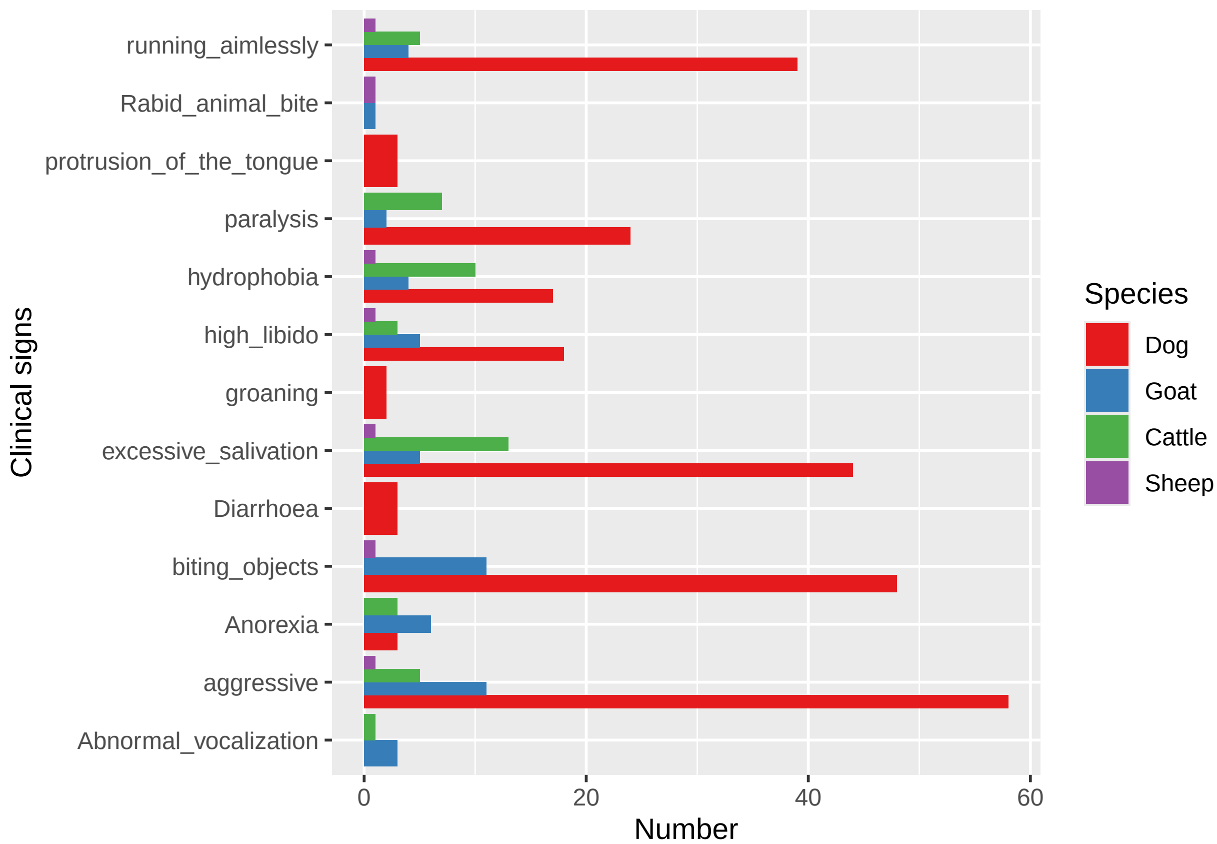
```
##                                              Estimate Std. Error z value
## (Intercept)                                   0.48530    1.04641   0.464
## aggressive                                    3.28508    1.63832   2.005
## biting_objects                               -0.07734    1.59228  -0.049
## excessive_salivation                          0.77180    1.09753   0.703
## aggressive:biting_objects                    -1.52643    0.96011  -1.590
## aggressive:excessive_salivation              -2.24131    1.06894  -2.097
## biting_objects:excessive_salivation           1.01704    1.30034   0.782
## aggressive:biting_objects:excessive_salivation  0.37675    0.29447   1.279
##                                              Pr(>|z|)
## (Intercept)                                    0.6428
## aggressive                                     0.0449 *
## biting_objects                                 0.9613
## excessive_salivation                           0.4819
## aggressive:biting_objects                      0.1119
## aggressive:excessive_salivation                0.0360 *
## biting_objects:excessive_salivation            0.4341
## aggressive:biting_objects:excessive_salivation   0.2008
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 68.275  on 81  degrees of freedom
## Residual deviance: 61.807  on 74  degrees of freedom
## AIC: 77.807
##
## Number of Fisher Scoring iterations: 8
```

```
##
## Call:
## glm(formula = lab_diagnostics_pcr ~ aggressive * excessive_salivation,
##     family = binomial(link = "logit"), data = model_glm)
##
## Coefficients:
##                                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)                      1.83785    0.84151   2.184    0.029 *
## aggressive                       0.21853    0.71909   0.304    0.761
## excessive_salivation            -0.38695    0.69668  -0.555    0.579
## aggressive:excessive_salivation  0.04118    0.21367   0.193    0.847
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 68.275  on 81  degrees of freedom
## Residual deviance: 67.696  on 78  degrees of freedom
## AIC: 75.696
##
## Number of Fisher Scoring iterations: 4
```

Time series




```
## Saving 6.5 x 4.5 in image
```

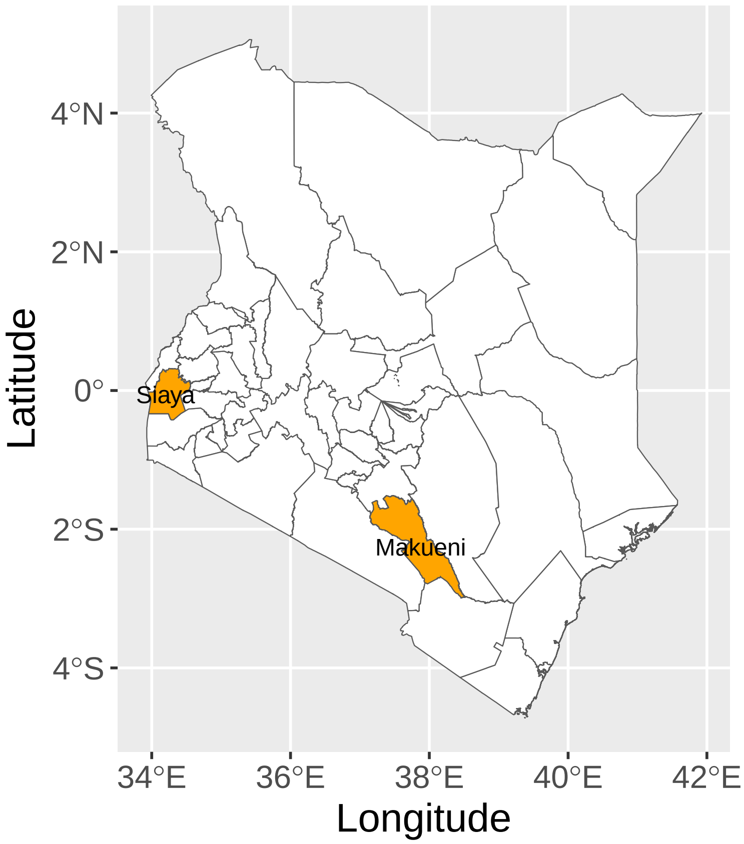



```
## Saving 6.5 x 4.5 in image
```

County plots

```r
ggplot()+
  geom_sf(data=counties, aes(geometry=geometry), fill="white")+
  geom_sf(data = chosen, aes(geometry=geometry),fill="orange")+
  geom_sf_text(data = chosen, aes(label = NAME_1),size = 3)+
  labs(x="Longitude",y="Latitude")+theme(text=element_text(size=15))
```

```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not
## give correct results for longitude/latitude data
```

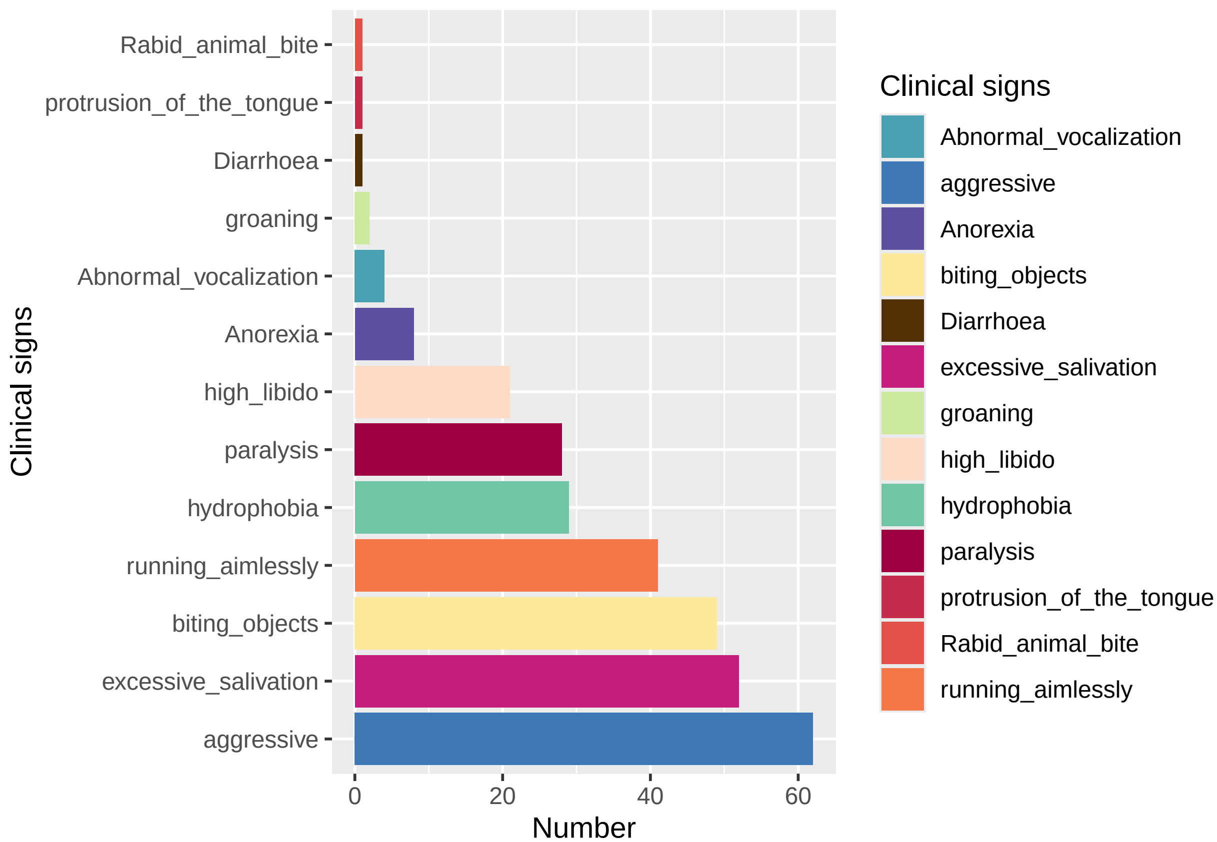


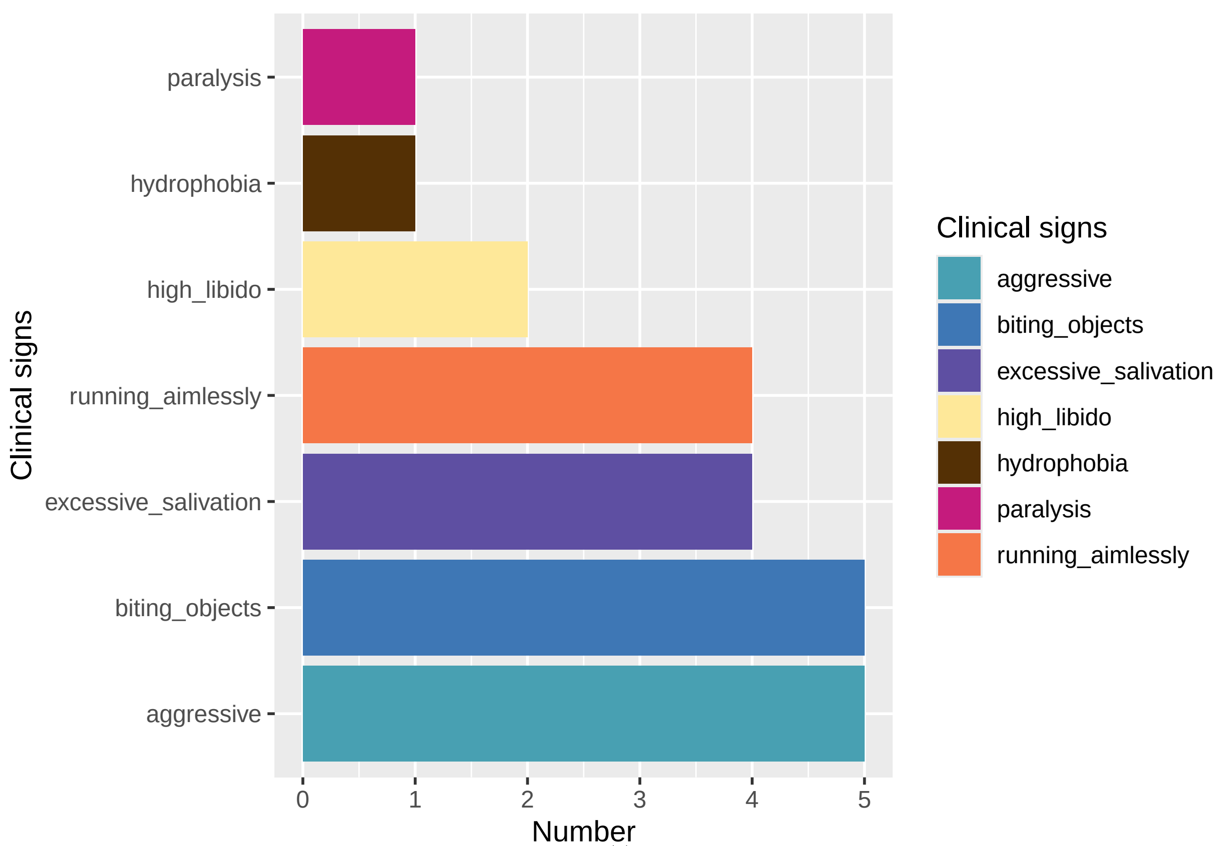

```r
ggsave("plot5.png")
```

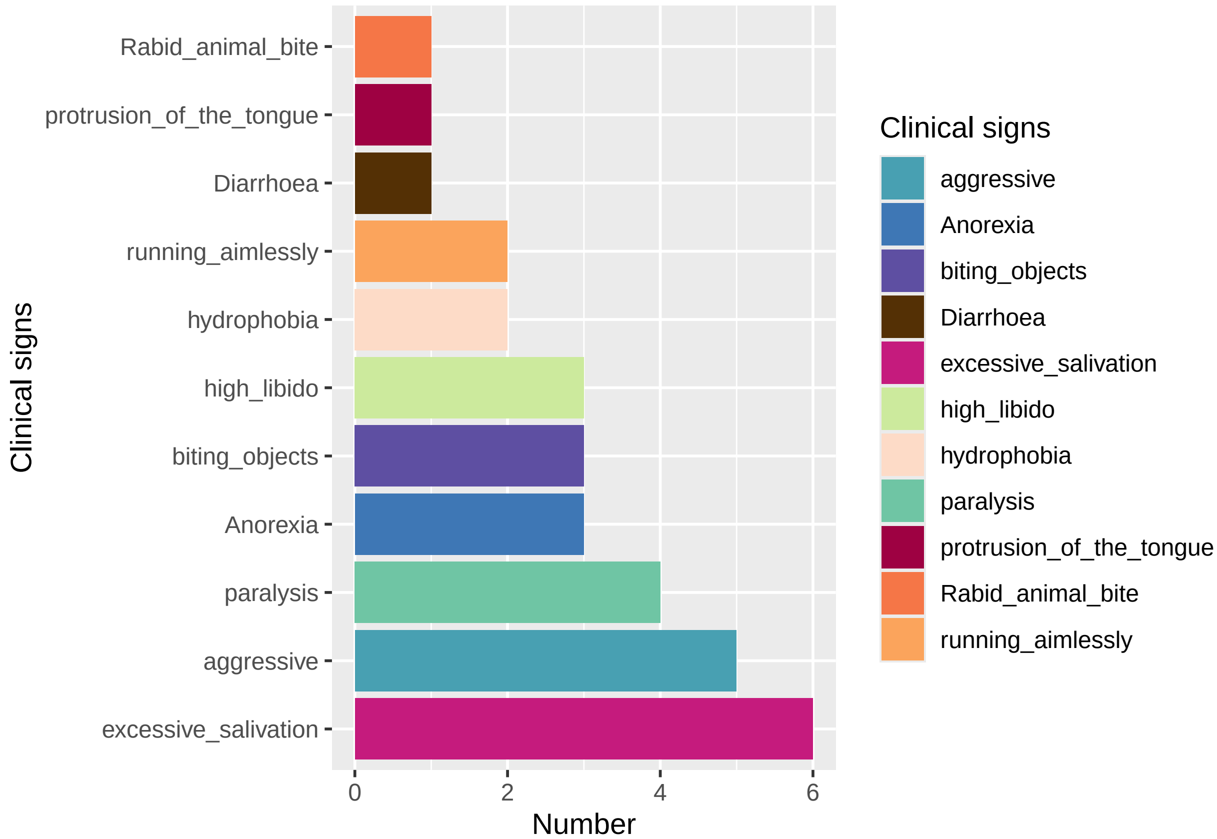
```
## Saving 6.5 x 4.5 in image
```

```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not
## give correct results for longitude/latitude data
```

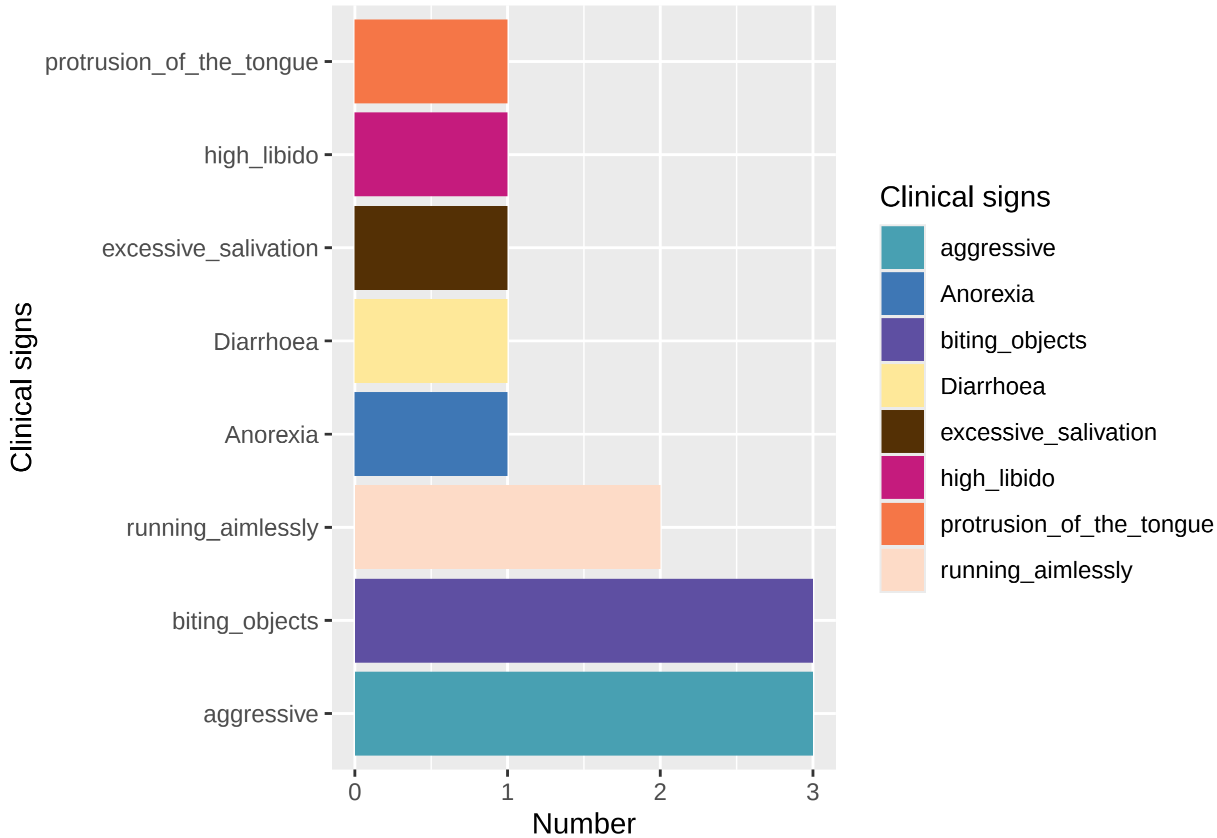
data with RDT & clinical signs







Clinical signs
Rabid_animal_bite
protrusion_of_the_tongue
Diarrhoea
running_aimlessly
hydrophobia
high_libido
biting_objects
Anorexia
paralysis
aggressive
excessive_salivation
0
2
4
6
Number
Clinical signs
aggressive
Anorexia
biting_objects
Diarrhoea
excessive_salivation
high_libido
hydrophobia
paralysis
protrusion_of_the_tongue
Rabid_animal_bite
running_aimlessly

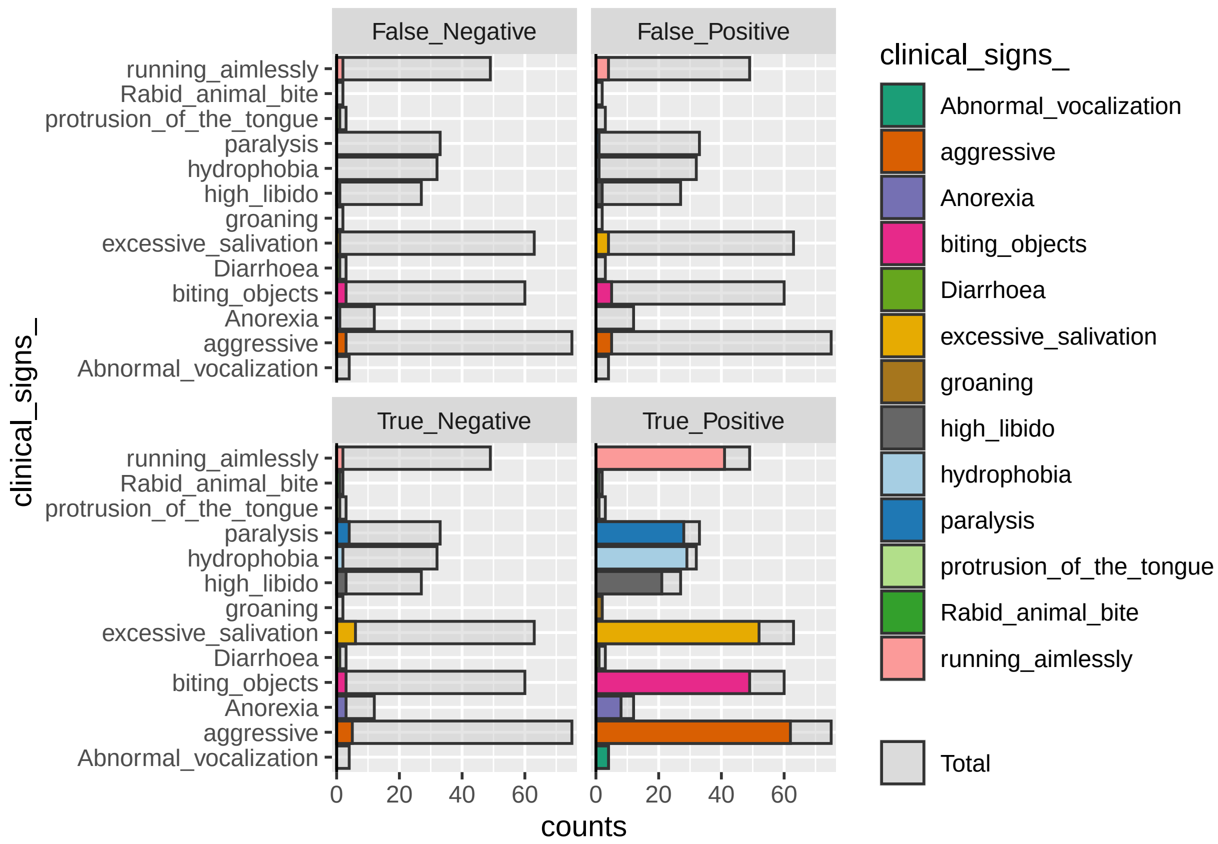



```
## Saving 6.5 x 4.5 in image
## Saving 6.5 x 4.5 in image
## Saving 6.5 x 4.5 in image
## Saving 6.5 x 4.5 in image
```

Pyramids

```
sample_data <- test |>
  select(clinical_signs_, test_result) |>
  group_by(clinical_signs_,test_result) |>
  dplyr::count()
  #dplyr::filter(test_result == "True_Positive" | test_result == "False_Positive")

sample_data <- as.data.frame(sample_data)

mycolors <- c(brewer.pal(name="Dark2", n = 8), brewer.pal(name="Paired", n = 6))

pr <- age_pyramid(sample_data,
  age_group = clinical_signs_,
  split_by = test_result,
  stack_by = clinical_signs_,
  count = n,
  na.rm = TRUE,
```

```
  show_midpoint = F,
  vertical_lines = F,
  horizontal_lines = F,
  pyramid = T,
  pal = mycolors
)

pr
```

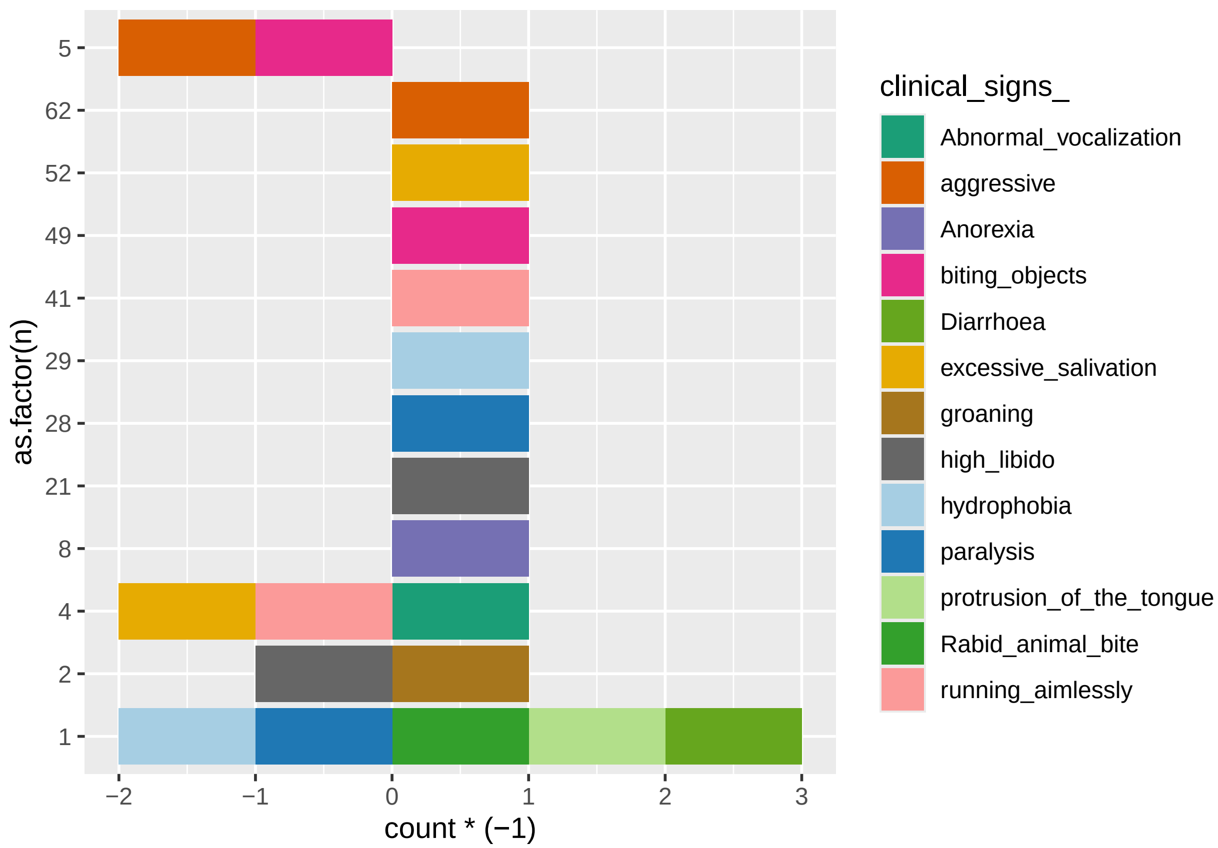



```
ggsave("pyramid.png", pr)
```

```
## Saving 6.5 x 4.5 in image
```

```
ggplot(data=sample_data,aes(x=as.factor(n),fill=clinical_signs_)) +
  geom_bar(data=subset(sample_data,test_result=="True_Positive")) +
  geom_bar(data=subset(sample_data,test_result=="False_Positive"),aes(y=..count..*(-1))) +
  scale_fill_manual(values = mycolors) +
  coord_flip()
```

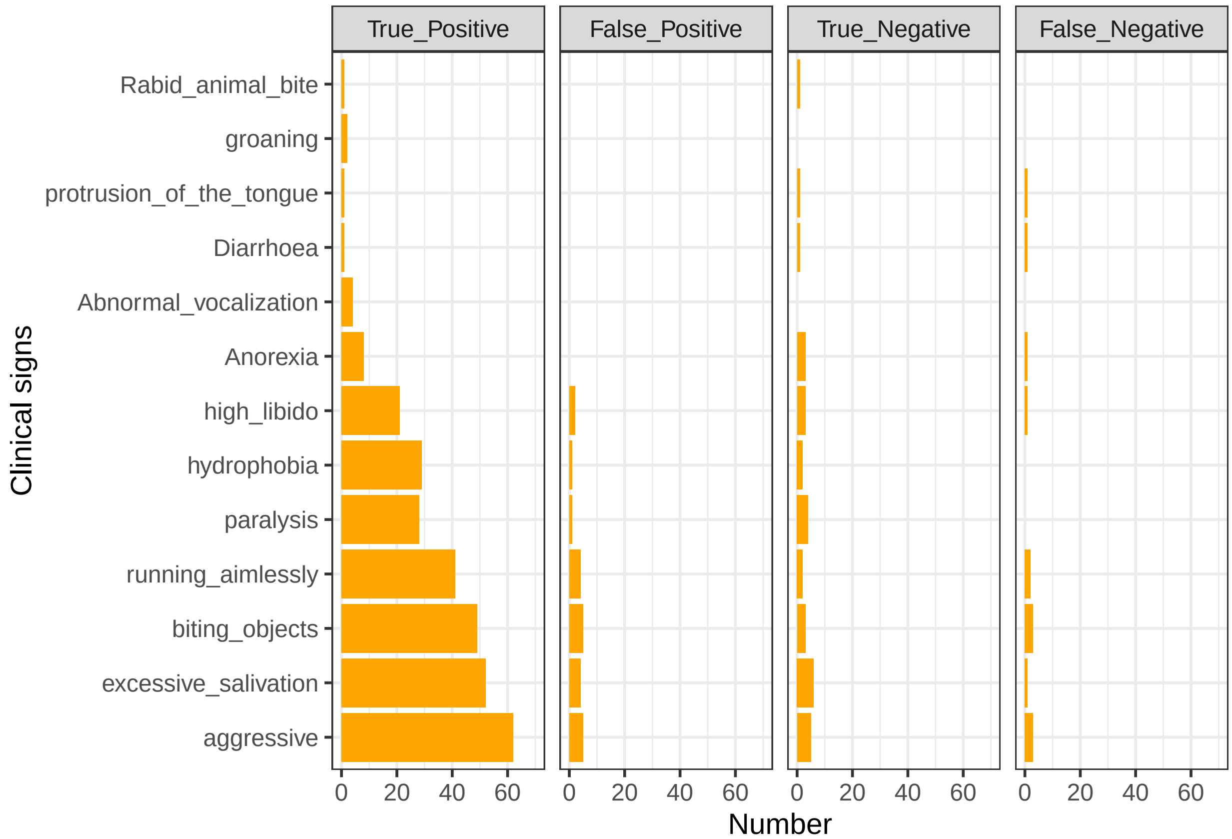
```
## Warning: The dot-dot notation (`..count..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(count)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



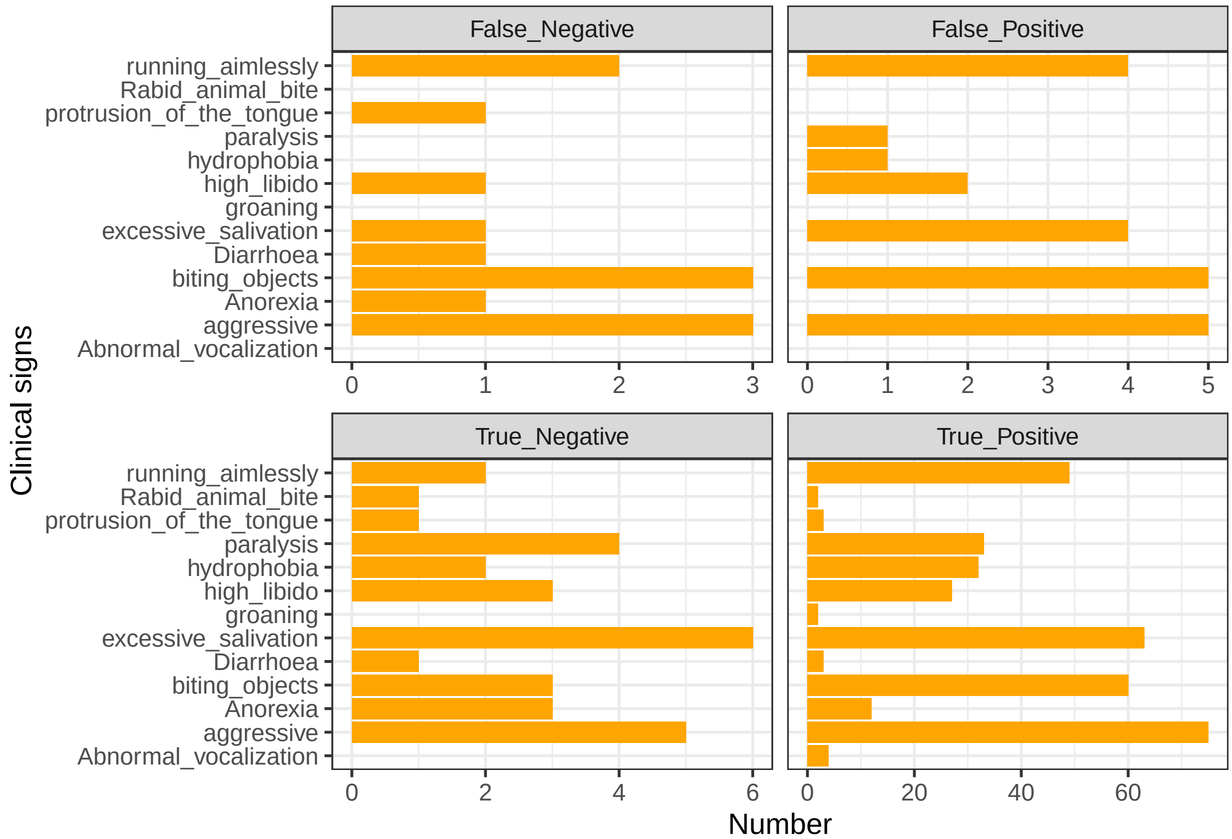

```
# library(ggplot2)
# library(lemon)
#
# ggplot(data = sample_data, mapping = aes(x = ifelse(test = test_result == "False_Positive" , yes = -t
#   geom_col() +
#   scale_x_symmetric(labels = abs) +
#   labs(x = "Population")
```

simple figure

```
sample_data$test_result <- factor(sample_data$test_result, levels = c("True_Positive", "False_Positive"
        "False_Negative" ))

ggplot(data = sample_data)+
  geom_col(aes(x=forcats::fct_infreq(clinical_signs_, n), y=n, fill = clinical_signs_), fill = "orange")
  labs(x = "Clinical signs", y = "Number", fill="Clinical signs")+
  facet_grid(.~test_result, scales = "free_x") +
  theme_bw()+
  ylim(c(0,70)) +
  coord_flip()
```




```r
ggplot()+
  geom_col(aes(x=n, y = clinical_signs_), fill = "orange", data = sample_data %>% mutate(subset = "True_
  geom_col(aes(x=n, y = clinical_signs_), fill = "orange", data = sample_data %>% filter(test_result ==
           mutate(subset = "False_Positive")) +
  facet_wrap(~ subset, scales = "free_x") +
  geom_col(aes(x=n, y = clinical_signs_), fill = "orange", data = sample_data %>% filter(test_result ==
           mutate(subset = "True_Negative")) +
  facet_wrap(~ subset, scales = "free_x") +
  geom_col(aes(x=n, y = clinical_signs_), fill = "orange", data = sample_data %>% filter(test_result ==
           mutate(subset = "False_Negative")) +
  facet_wrap(~ subset, scales = "free_x") +
  labs(y = "Clinical signs", x = "Number") +
  theme_bw()
```

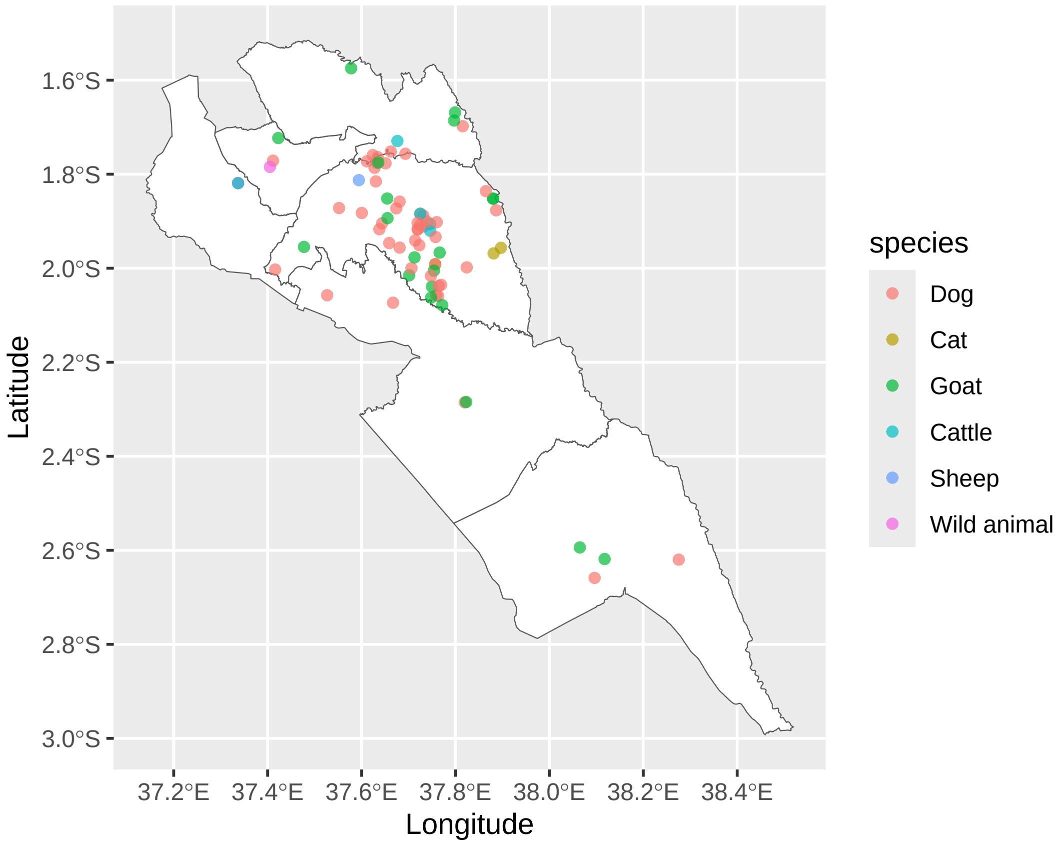



```r
ggsave("plot6.png")
```

```
## Saving 6.5 x 4.5 in image
```

Mapping animal samples

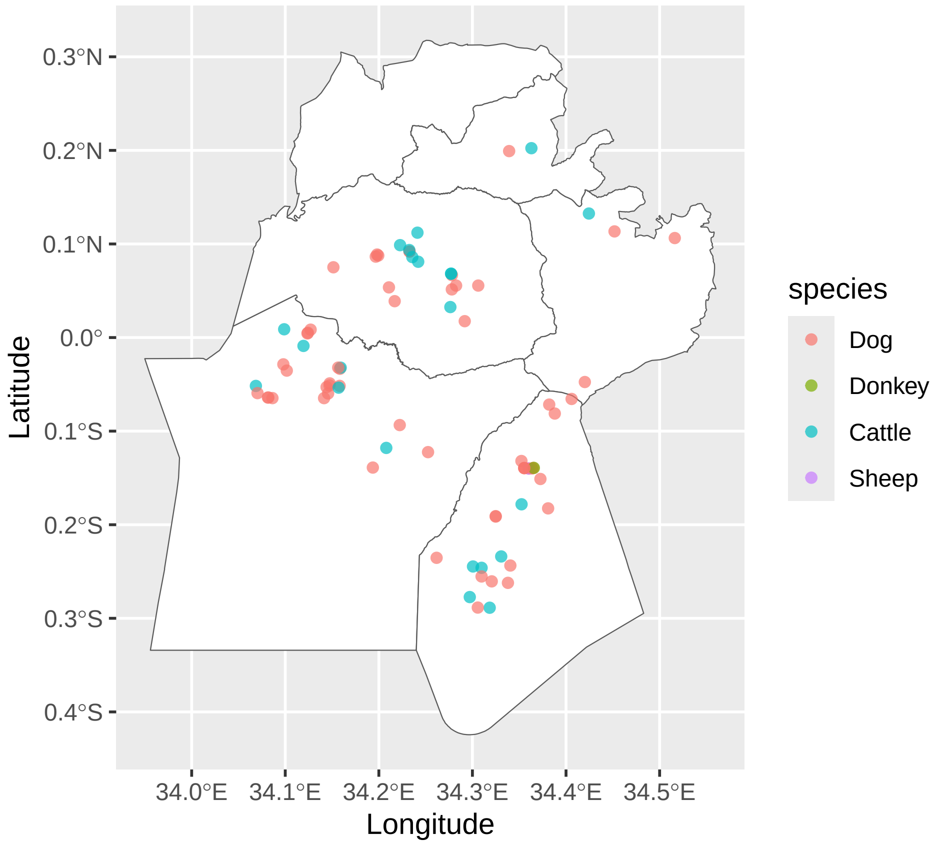
```r
# Makueni

ggplot()+
  geom_sf(data=makueni_map, aes(geometry=geometry), fill = "white")+
  geom_sf(data = makueni_sd, aes(geometry = geometry, col = species),alpha = 0.7)+
  #theme_bw()+
  labs(x="Longitude",y="Latitude")
```




```
  #theme(text=element_text(size=18))

ggsave("plot7.png")
```

```
## Saving 6.5 x 4.5 in image
```

```
# Siaya

ggplot()+
  geom_sf(data=siaya_map, aes(geometry=geometry), fill = "white")+
  geom_sf(data = siaya_sd, aes(geometry = geometry, col = species),alpha = 0.7)+
  #theme_bw()+
  labs(x="Longitude",y="Latitude")
```




```
  #theme(text=element_text(size=18))

ggsave("plot8.png")
```

```
## Saving 6.5 x 4.5 in image
```